

EYITAYO EZEKIEL

BACKEND ENGINEER • DISTRIBUTED SYSTEMS • TRANSACTIONAL ARCHITECTURE

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Portfolio: <https://eyitayo.online> | **GitHub:** <https://github.com/dev-eyitayo>

PROFESSIONAL SUMMARY

Backend engineer with production experience in transactional systems, API design, and performance debugging. Background includes integrating third-party payment and BNPL services into multi-channel business platforms, building reliable healthcare infrastructure, and developing retrieval systems evaluated with real information retrieval metrics. Comfortable working across high-level API design, database schema planning, and low-level debugging of concurrency, state, and data consistency issues.

WORK EXPERIENCE

Backend Engineer — Sync360 <i>Fintech & Multi-Channel Business Management Platform</i>	May 2026 – Present Remote
- Integrated an external Buy-Now-Pay-Later (BNPL) service into the core checkout system, designing transactional state handling, webhook listeners, and retry queues to manage partial payments and edge cases cleanly. - Architected and maintained RESTful APIs for order routing, sales tracking, and a multi-tier customer loyalty rewards system, maintaining 99.9% uptime across high transaction volumes. - Profiled and optimized database queries using composite indexes and execution plan analysis, cutting p95 query latency by 42% on high-traffic sales endpoints. - Built automated asynchronous ETL reporting pipelines using Celery and openpyxl, pulling sales, expense, and inventory data into scheduled monthly Excel reports and eliminating 25+ hours of manual bookkeeping weekly. - Built a tool-calling AI Business Insights assistant over database aggregation queries, letting business owners query sales trends and inventory status in plain language with strict tenant data isolation. - Set up automated CI/CD test workflows with GitHub Actions and Docker, raising backend test coverage from 60% to 88% with pytest and standardizing OpenAPI documentation.	
Intern Technologist — Blueapril Technologies <i>Enterprise Cloud Solutions & Software Consultancy</i>	Oct 2025 – April 2026 Remote
- Participated in system architecture and schema design for a multi-tenant cooperative management platform, evaluating data isolation strategies and using PostgreSQL Row-Level Security (RLS) to enforce tenant boundaries. - Built internal administrative dashboards and REST endpoints to support tenant onboarding and day-to-day operations, reducing onboarding turnaround time by 40%. - Conducted peer code reviews across 40+ pull requests, helping enforce consistent error handling, PEP8 standards, and clean architecture practices.	
Backend Engineer — Toltimedcare <i>Digital Healthcare & Telemedicine Platform</i>	Aug 2024 – Feb 2026 Remote
- Maintained backend uptime (99.9%) and reliability for core healthcare workflows, including patient appointment booking, telehealth consultation routing, and medical records. - Designed secure REST APIs with database and caching layers, implementing RBAC authentication and AES-256 field encryption for sensitive patient health data. - Diagnosed and resolved database N+1 query bottlenecks and connection pool contention, reducing p95 response times from 1.3s to under 220ms under concurrent loads. - Implemented structured logging across core services, providing actionable error context and dramatically reducing time spent reproducing and resolving production bugs.	

TECHNICAL PROJECTS & RESEARCH

Sheltly — AI Semantic Search & Explainable Ranking — Undergrad Research

Final-Year Thesis | College

Technologies: Python, PostgreSQL (pgvector), Sentence-BERT, RankSHAP, Redis

- Built a real-estate search backend using Sentence-BERT (all-MiniLM-L6-v2, 384 dimensions) embeddings and PostgreSQL pgvector cosine similarity to match listings by intent rather than exact keyword matches.
- Implemented an explainable ranking engine with SHAP feature attributions (/ai/explain), allowing users and researchers to see how price, recency, location distance, and semantic similarity contribute to listing rank.
- Evaluated retrieval performance quantitatively against ground-truth benchmarks, achieving 0.84 NDCG@10, 0.79 MAP, and 0.88 MRR compared to PostgreSQL full-text search (tsvector).
- Added Haversine radius filtering and Redis caching for geocoding lookups, keeping search response times under 80ms across 10,000+ property records.

Kontributa — Automated Fintech Contribution Platform — Personal Project

Fintech Platform | Production

Technologies: Python, PostgreSQL, Redis, Alembic, Monnify API, Paystack API, Docker

- Built a fintech backend enabling community groups (associations, cooperatives, religious bodies) to collect member contributions with automated bank transfer verification via Monnify and Paystack dynamic virtual accounts.
- Built idempotent webhook consumers with Redis distributed locking and exponential backoff, preventing duplicate transaction processing and eliminating manual payment receipt verification.
- Implemented backend logic for partial payments, manual payment logging, and audit-logged ledgers with double-entry accounting models.

CLI & Systems Tool | Linux

Kitsune — Linux Web App Isolation & Desktop Manager — Open-Source Tool

Technologies: Python (Standard Library), Linux (X11/Wayland), Firefox Gecko, IPC

- Built a zero-dependency Linux CLI and desktop integration tool that transforms web applications into isolated, standalone desktop apps powered by the Firefox Gecko engine.
- Implemented process sandboxing and session isolation by generating dedicated Firefox profiles with isolated SQLite cookie stores, caches, and unique MOZ_APP_REMOTINGNAME IPC channels to prevent instance hijacking.
- Solved Wayland/GNOME dock window-grouping conflicts using XWayland window class targeting (StartupWMClass), generating native Freedesktop .desktop entries and injecting custom userChrome.css for frameless window rendering.
- Engineered a native messaging bridge to automatically detect and route external hyperlink navigation from the isolated web app back to the user's primary browser instance.

EDUCATION

Bachelor of Science in Software Engineering — Lead City University

Sep 2022 – July 2026 | Ibadan, Nigeria

Relevant Coursework: Data Structures & Algorithms, Distributed Systems, Database Management Systems, Software Architecture, Machine Learning, Information Retrieval, Cloud Computing

TECHNICAL SKILLS

- **Languages:** Python (3.10+), JavaScript (ES6+), TypeScript, SQL, Bash
- **Frameworks & Tools:** FastAPI, Django REST Framework, Node.js, Nest.js, Express.js, Celery, openpyxl, Pydantic, SQLAlchemy, Next.js, React.js
- **Databases & Caching:** PostgreSQL (with pgvector), Redis, MongoDB, Firebase Firestore
- **Architecture & Systems:** RESTful API Design, Transaction Management & Idempotency, Multi-Tenancy (RLS), Queue-Based Architecture, Vector Similarity Search, Database Indexing & Query Tuning
- **DevOps & Cloud:** Docker, Git/GitHub, CI/CD (GitHub Actions), AWS (S3, EC2), Linux, Sentry, Swagger/OpenAPI, Postman